

APPENDIX A REVISITING CRITICAL PERIODS

In this experiment, we produce 200 oracle models and Figure 1 shows the accuracy of each one at the end of the training phase. Figure 1 reinforces the existence of critical periods early on training and supports that, after pinpointing i^* (i.e. the end of the critical period), reducing augmentation is optimal in terms of accuracy and computational cost. Following this reasoning, epoch 24 marks the potential end of the critical period. It is important to note that better-performing models only emerge by epoch 60, but the accuracy gain is not significant enough to justify the additional computational cost.

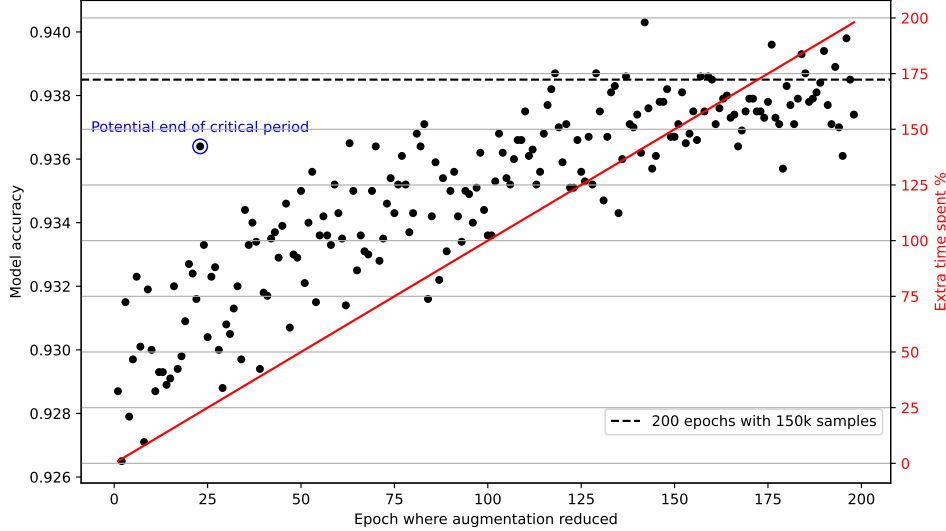


Fig. 1. Critical period in ResNet32 training on CIFAR-10. Each point indicates a model with a complete training cycle of 200 epochs. The figure shows the accuracy trends when data augmentation reduces from 150k to 50k samples per epoch at different epochs. The red line indicates the extra time cost.

APPENDIX B COMPARISON WITH STATE-OF-THE-ART DATA PRUNING

In this experiment, we investigate the behavior of applying data pruning *only after* our method identifies critical periods and compare the results with state-of-the-art methods. For this purpose, we build upon the experimental framework established by [41], using their evaluation of static and dynamic data pruning methods* as a benchmark. To this baseline, we integrate results from our approach, introducing data pruning after identifying the critical periods during training. For a fair comparison, we use ResNet18 as the backbone model, training it on CIFAR-10/100 datasets with pruning ratios of 30%, 50% and 70%. It is worth mentioning our method diverges from traditional data pruning techniques by initially training the model on the full dataset (i.e $k = 1$) for the first few epochs and then applying the pruning ratio from the critical periods onward. We compute the accuracy delta relative to a model trained for 200 epochs without data pruning, ensuring a fair comparison. Table I shows the results.

According to Table I, our method outperforms all static pruning methods across both datasets and all pruning ratios, consistently achieving lower accuracy degradation. Compared to dynamic methods, our approach delivers competitive performance, rivaling the state-of-the-art InfoBatch method [41]. For example, on CIFAR-10 with a 50% pruning ratio, our method results in an accuracy drop of only 0.3. Additionally, on CIFAR-100 with a 30% pruning ratio, our method leads to a surprising increase in accuracy by 0.2, showcasing its robustness.

Overall, the previous results emphasize the importance of timing in data pruning, demonstrating that applying these mechanisms after critical periods allows the model to retain more data during the most effective learning phases; thereby optimizing performance while reducing trade-offs.

APPENDIX C EFFECTIVENESS ON LARGE DATASETS AND OPTIMIZERS.

To validate the effectiveness of our method, we analyze its performance across different datasets and optimizers. Our objective is to demonstrate that the method is agnostic to both dataset and optimizer choices, further reinforcing its general applicability.

*Static data pruning removes a predetermined subset of data, with no reintroduction during training, whereas dynamic pruning allows for the subset to change as training progresses, adapting to the requirements of the model [41]. Our method, as specified in Section III, uses the dynamic approach.

TABLE I
COMPARISON OF ACCURACY DELTA (ΔAcc) WITH STATE-OF-THE-ART DATA PRUNING METHODS. BOLD, UNDERLINE, $\uparrow$ AND $\downarrow$ MEAN THE BEST RESULTS, SECOND-BEST RESULTS, ACCURACY IMPROVEMENT AND ACCURACY DECREASE, RESPECTIVELY.

Dataset		CIFAR-10			CIFAR-100		
Pruning Ratio %		30	50	70	30	50	70
Static	Random	$\downarrow 1.0$	$\downarrow 2.3$	$\downarrow 5.4$	$\downarrow 4.4$	$\downarrow 6.1$	$\downarrow 8.5$
	CD	$\downarrow 0.6$	$\downarrow 1.3$	$\downarrow 4.8$	$\downarrow 4.0$	$\downarrow 5.9$	$\downarrow 7.9$
	Herding	$\downarrow 3.4$	$\downarrow 7.6$	$\downarrow 15.5$	$\downarrow 5.1$	$\downarrow 6.4$	$\downarrow 8.0$
	K-Center	$\downarrow 0.9$	$\downarrow 1.7$	$\downarrow 4.7$	$\downarrow 4.1$	$\downarrow 6.0$	$\downarrow 8.0$
	Least Confidence	$\downarrow 0.6$	$\downarrow 1.1$	$\downarrow 5.3$	$\downarrow 4.0$	$\downarrow 5.9$	$\downarrow 8.4$
	Margin	$\downarrow 0.7$	$\downarrow 1.3$	$\downarrow 4.7$	$\downarrow 4.2$	$\downarrow 6.0$	$\downarrow 8.0$
	Forgetting	$\downarrow 0.9$	$\downarrow 1.5$	$\downarrow 3.9$	$\downarrow 2.9$	$\downarrow 5.1$	$\downarrow 8.3$
	GraNd-4	$\downarrow 0.3$	$\downarrow 1.0$	$\downarrow 4.4$	$\downarrow 3.6$	$\downarrow 6.8$	$\downarrow 9.4$
	DeepFool	$\downarrow 0.5$	$\downarrow 1.5$	$\downarrow 5.6$	$\downarrow 4.0$	$\downarrow 5.0$	$\downarrow 6.4$
	Craig	$\downarrow 0.8$	$\downarrow 3.3$	$\downarrow 7.2$	$\downarrow 3.8$	$\downarrow 6.3$	$\downarrow 8.5$
	Glistner	$\downarrow 0.4$	$\downarrow 1.6$	$\downarrow 4.7$	$\downarrow 3.6$	$\downarrow 5.0$	$\downarrow 7.8$
	Influence	$\downarrow 2.5$	$\downarrow 4.3$	$\downarrow 7.3$	$\downarrow 3.8$	$\downarrow 6.2$	$\downarrow 9.5$
	EL2N-2	$\downarrow 1.2$	$\downarrow 2.4$	$\downarrow 5.8$	$\downarrow 4.1$	$\downarrow 7.2$	$\downarrow 9.7$
	EL2N-20	$\downarrow 0.3$	$\downarrow 0.5$	$\downarrow 3.7$	$\downarrow 1.0$	$\downarrow 6.1$	-
	DP	$\downarrow 0.7$	$\downarrow 1.8$	$\downarrow 4.8$	$\downarrow 1.0$	$\downarrow 5.1$	-
Dynamic	Random	$\downarrow 0.8$	$\downarrow 1.1$	$\downarrow 2.6$	$\downarrow 0.9$	$\downarrow 2.9$	-
	ϵ -greedy	$\downarrow 0.4$	$\downarrow 0.7$	$\downarrow 1.5$	$\downarrow 1.8$	$\downarrow 3.4$	-
	UCB	$\downarrow 0.3$	$\downarrow 0.9$	$\downarrow 1.7$	$\downarrow 0.9$	$\downarrow 2.9$	-
	InfoBatch	$\uparrow 0.0$	$\downarrow 0.5$	$\downarrow 0.9$	$\uparrow 0.0$	$\downarrow 0.1$	$\downarrow 1.7$
	Ours	$\uparrow 0.0$	$\downarrow 0.3$	$\downarrow 1.6$	$\uparrow 0.2$	$\downarrow 1.4$	$\downarrow 1.8$

To this end, we consider the following datasets: CIFAR-10/100 [6], [7], EuroSAT [8], TinyImageNet [9] and ImageNet30 [10]. On these datasets, the popular architecture is ResNet50; therefore, we employ it in these experiments.

Table II introduces the accuracy delta and time saved when applying our method on large datasets. The results confirm that our approach maintains competitive performance while significantly reducing computational costs, independent of the dataset. More concretely, we save up to 65.79% of training time while increasing accuracy on EuroSAT by 1.02 pp.

TABLE II
ACCURACY DELTA (ΔAcc) IN PERCENTAGE POINTS AND TIME SAVED OF OUR METHOD ACROSS DIFFERENT DATASETS USING RESNET50 WITH AUGMENTATION FACTOR $k = 3$. $\uparrow$ AND $\downarrow$ MEAN ACCURACY IMPROVEMENT AND DECREASE, RESPECTIVELY.

	CIFAR-10	CIFAR-100	EuroSAT	TinyImageNet	ImageNet30
ΔAcc	$\downarrow 0.32$	$\downarrow 2.36$	$\uparrow 1.02$	$\downarrow 0.20$	$\uparrow 0.34$
Saved (%)	49.35	49.47	65.79	60.31	61.70

Similarly, Table III examines the impact of different optimizers on the effectiveness of our method in identifying critical periods. This experiment is important to show that our method is neither confined nor biased to the iterative learning process of SGD. Table III confirms the consistency of our method across different optimizers, demonstrating its optimizer-agnostic nature and ensuring applicability in diverse training settings. The results indicate that changing the optimizer has minimal impact on our method. The accuracy deltas remain within a narrow range, and the time saved is consistently around 33% across all optimizers, reinforcing the robustness of our approach.

TABLE III
ACCURACY DELTA (ΔAcc) IN PERCENTAGE POINTS AND TIME SAVED OF OUR METHOD ACROSS DIFFERENT OPTIMIZERS USING RESNET50 ON CIFAR-10 WITH AUGMENTATION FACTOR $k = 2$. $\uparrow$ AND $\downarrow$ MEAN ACCURACY IMPROVEMENT AND DECREASE, RESPECTIVELY.

	SGD	AdamW	RMSprop	AdaGrad
ΔAcc	$\downarrow 1.04$	$\downarrow 0.18$	$\downarrow 0.59$	$\downarrow 0.32$
Saved (%)	33.75	33.68	32.99	33.57

These results, combined with our previous experiments on different architectures, reinforce that our method is not only model-agnostic, as previously demonstrated, but also dataset-agnostic and optimizer-agnostic, further supporting its wide-ranging applicability.